



Efficacy and safety of Janus kinase inhibitors in systemic and cutaneous lupus erythematosus: A systematic review and meta-analysis

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ABSTRACT

Background: Janus kinase (JAK) inhibitors have been proven to be effective and safe in various autoimmune diseases. However, there is still a lack of comprehensive evidence regarding their efficacy and safety in systemic and cutaneous lupus erythematosus.

Methods: We searched for systemic and cutaneous lupus erythematosus patients who were treated with JAK inhibitors in PubMed, Embase, Web of Science, and the Cochrane Library until February 28, 2023. The quality of clinical trials was assessed using the Cochrane risk-of-bias tool. Meta-analysis was conducted when at least three studies had comparable measures of outcome. If meta-analysis was not feasible, a descriptive review was carried out.

Results: We included 30 studies, consisting of 10 randomized controlled trials and 20 case series or reports, with a total of 2,460 patients. JAK inhibitors were found to be more effective than placebo in systemic lupus erythematosus (SLE) based on the percentage of achieving SLE Responder Index (SRI)-4 response (RR = 1.18; 95% CI 1.07 to 1.31; $p = 0.001$), British Isles Lupus Assessment Group -based Composite Lupus Assessment (BICLA) response (RR = 1.16; 95% CI 1.02 to 1.31; $p = 0.02$), Lupus Low Disease Activity State (LLDAS) (RR = 1.28; 95% CI 1.07 to 1.54; $p = 0.008$), and Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2 K) remission of arthritis or rash (RR = 1.09; 95% CI 1.00 to 1.18; $p = 0.04$), particularly in treating musculoskeletal and mucocutaneous involvement. However, the effect of JAK inhibitors on cutaneous lupus erythematosus was uncertain. JAK inhibitors and placebo had a similar incidence of adverse events (RR = 1.01; 95% CI 0.97 to 1.04; $p = 0.65$).

Conclusion: JAK inhibitors could be a potential treatment option for systemic and cutaneous lupus erythematosus, particularly in treating cutaneous and musculoskeletal lesions of SLE. JAK inhibitors had a safe profile.

1. Introduction

Lupus is an autoimmune connective tissue disease characterized by autoantibodies and immune complex formation [1]. Systemic lupus erythematosus (SLE) involves multiple organs, while cutaneous lupus erythematosus (CLE) presents as an independent skin involvement or as a manifestation of SLE [2]. Established therapies are broad-spectrum,

including glucocorticoids and immunosuppressive agents that are counterbalanced by their side effects [3]. Even though emerging therapies are burgeoning, represented by biological disease-modifying antirheumatic drugs including belimumab, anifrolumab, and telitacicept, we are still unable to radically alter the course of these diseases [4]. Therefore, more effective, safe, and targeted therapies are needed.

To date, JAK inhibitors have been approved for use in rheumatoid

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arthritis, psoriatic arthritis, and ankylosing spondylitis [5]. In recent years, scientists have started to explore their application in lupus. JAK has four isoforms, JAK1, JAK2, JAK3, and tyrosine kinase 2 (TYK2). Correspondingly, the first-generation pan-JAK inhibitors targeted two or more isoforms, while the second-generation selective JAK inhibitors targeted one specific isoform. In recent years, clinical trials of various JAK inhibitors, such as baricitinib and filgotinib in SLE or CLE were blossoming, but most of them remained in phase I and II stages, with no indications of lupus being approved yet. On February 24, 2023, results of two phase III RCTs of baricitinib for SLE (SLE-BRAVE-I and II) were published [6,7], highlighting this as a hot spot. On the other hand, case reports and series that clarified the successful application of JAK inhibitors in SLE and CLE continued to come out, calling out the urgent demands for scientific evidence of this practice. To fill this gap, we conducted a systematic review to evaluate the efficacy and safety of JAK inhibitors for lupus and to determine their pooled effectiveness.

2. Methods

2.1. Search strategy

We conducted a systematic review following the PRISMA guidelines by searching PubMed, Embase, Web of Science and the Cochrane Library for studies on SLE and CLE treated with JAK inhibitors until February 28, 2023. We used terms related to JAK inhibitors and lupus to capture relevant articles. The complete search strategies were presented in Supplementary Table S1. References of relevant articles were manually identified and reviewed. We also searched for clinical trials on [Clinicaltrials.gov](https://clinicaltrials.gov) to include any relevant data that may have been missing from the previous database search. Two authors conducted the search independently and disagreements were resolved through consensus or by a third party.

2.2. Inclusion and exclusion criteria

We searched the literature for clinical trials, case reports, case series, and observational studies. Only the most comprehensive and recent piece in a series of articles reporting the same RCT was included. The studies were restricted to English publications. Reviews, preclinical studies, conference abstracts, expert opinions, editorials, comments, and repeated publications were excluded. We also excluded studies that did not have efficacy and safety outcomes (Supplementary Table S2).

2.3. Data extraction

Two authors independently extracted data from the included studies, including the year of publication, first and corresponding author, study type, disease type, age, gender, disease duration (before the initiation of JAK inhibitors), previous medications, JAK inhibitor type, dosage, administration, onset time and duration, disease activity score, clinical features, laboratory tests, glucocorticoids reduction, and adverse events. We used a predefined extraction form to collect data, which is presented in Supplementary Table S3.

2.4. Risk of bias

The quality of RCTs was assessed by the Cochrane risk-of-bias tool [8]. The level of evidence was determined by the Oxford Center for Evidence-Based Medicine (OCEBM) tool [9].

2.5. Statistical analysis

Meta-analysis was conducted when at least three low or moderate risk of bias studies had comparable measures of outcome. If meta-analysis was not feasible, a descriptive review was carried out. Statistical analyses were performed using Review Manager (RevMan) v5.4

(the Cochrane Collaboration, Copenhagen, Denmark). We analyzed dichotomous data using a random-effects model, and calculated relative risk (RR) with 95% confidence intervals (CI) [10]. The risk of publication bias was assessed visually by funnel plots symmetry (Supplementary Fig S1) [11]. Heterogeneity was tested using I^2 , with values >50% indicating heterogeneity among studies [12]. A p -value of <0.05 was considered statistically significant. Sensitivity analysis was performed by removing one single study at a time to evaluate its impact on the overall results.

3. Results

3.1. Study included

A flowchart outlining the search and selection process is shown in Fig. 1. After removing duplicates, a total of 724 articles were screened, and 605 of them were excluded due to irrelevance or being reviews. Of the remaining 119 studies, after full-text reading, 89 were excluded because they were preclinical studies, excluded literature styles, or had insufficient results. In the end, 30 studies comprising 2,460 patients (2,318 SLE patients and 142 CLE patients) were selected, including ten RCTs (seven for SLE and three for CLE) and 20 case reports and series (eight for SLE and twelve for CLE).

The characteristics of the included studies are summarized in Table 1. Among the seven RCTs of SLE, three compared baricitinib, one compared deucravacitinib, one compared tofacitinib, one compared GSK2586184 with placebo, respectively, and the left one compared filgotinib with lanraplenib. Three RCTs of CLE compared topical R333 with placebo, topical delgocitinib with placebo, and filgotinib with lanraplenib. Baricitinib, tofacitinib, and ruxolitinib were the most frequently used JAK inhibitors in case reports and series. Two of the included RCTs were deemed to have a high risk of bias due to a notable proportion of missing outcomes, which was significant enough to potentially affect the accuracy of the study results (Fig. 2) [8]. The level of evidence of all included studies is presented in Table 1 [9].

3.2. Patient characteristics at baseline

The baseline characteristics of patients are shown in Supplementary Tables S4 and S5.

A total of 2318 patients were included in the studies of SLE, with a mean age of 42.0 years, and all were adults. Females were the majority, accounting for 93.7%. The mean disease duration was 9.2 years. Most patients had mild to moderate disease activity, and the baseline mean Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2 K) scores of RCTs were similar, indicating consistency among studies. As the cornerstones of SLE treatment, over 70% of patients were given systemic glucocorticoids and antimalarials at baseline, and half of them were also concomitantly treated with immunosuppressors. The most commonly involved organs were the mucocutaneous and musculoskeletal systems, both of which were present in about 90% of patients. Renal damage and hematological involvement were relatively rare.

A total of 142 patients were included in the studies of CLE, with a mean age of 46.7 years, and all were adults except for one 3-year-old child with chilblain lupus erythematosus (ChLE). Females accounted for 81.0% of the patients. The disease duration varied, with an average of 9.7 years. Of the three included RCTs, two were for discoid lupus erythematosus (DLE), and one for subacute cutaneous lupus erythematosus (SCLE) or chronic cutaneous lupus erythematosus (CCLE). Case reports and series included five DLE, four SCLE, four ChLE, and one blaschko linear lupus erythematosus (BLLE). All patients had at least one active skin lesion, and the majority of them had tried various treatments such as increasing the glucocorticoid dose, applying topical glucocorticoids, and adding additional immunosuppressors, but none had satisfactory effects.

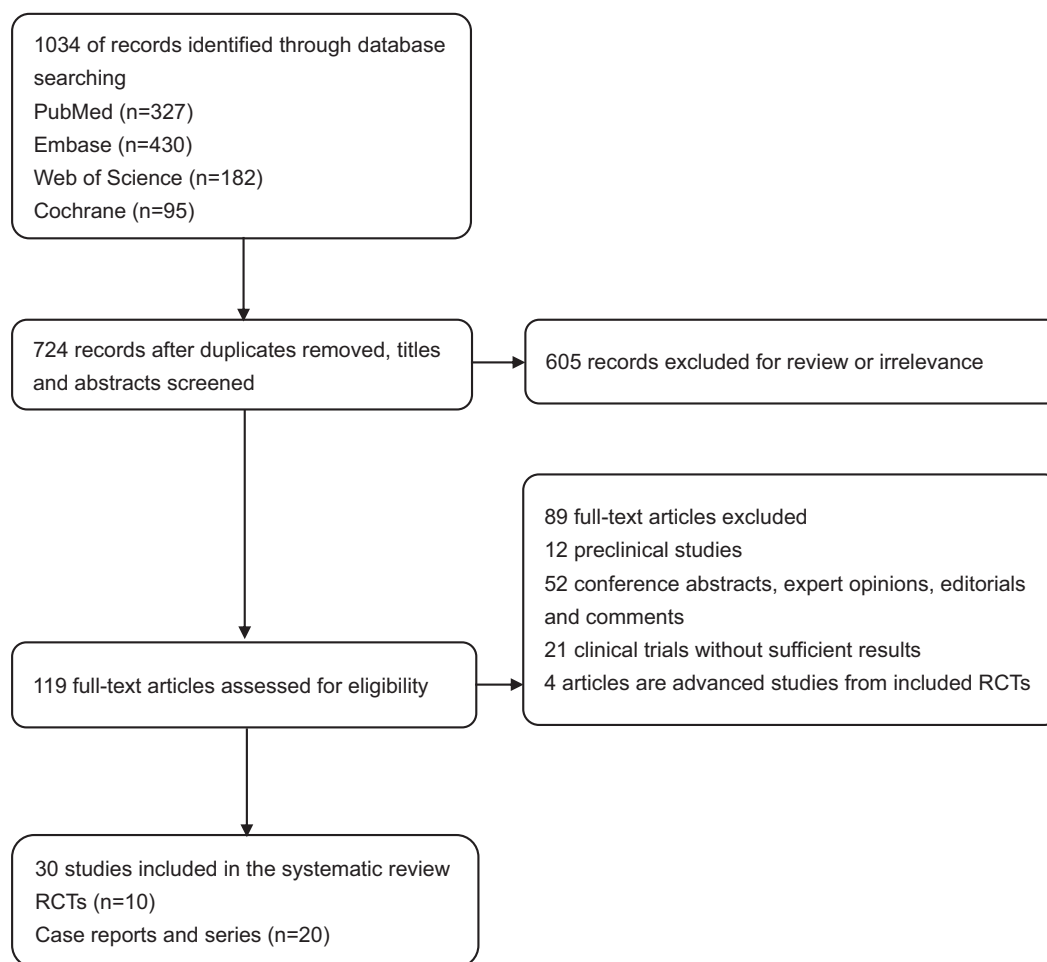


Fig. 1. Flow diagram of the search and selection process. RCTs, randomized controlled trials.

3.3. Effects of JAK inhibitors on SLE activity

Seven RCTs reported disease activity assessments before and after treatment. Combined analysis of the Systemic Lupus Erythematosus Responder Index-4 (SRI-4) achieved percentage yielded an RR of 1.18 (95% CI 1.07 to 1.31; $p = 0.001$, Fig. 3a) for patients using JAK inhibitors compared with placebo treatment. Sensitivity analyses showed the robustness of the synthesized result. We performed separate evaluations on three RCTs for baricitinib, and the SRI-4 united analysis showed an RR of 1.12 (95% CI 1.03 to 1.22; $p = 0.01$, Fig. 3b). After removing the deucravacitinib study, the heterogeneity was much lower ($I^2 = 7\%$). Furthermore, as shown in Supplementary Fig S1b, the publication bias is small. The British Isles Lupus Assessment Group (BILAG)-based Composite Lupus Assessment (BICLA) response percentage and Lupus Low Disease Activity State (LLDAS) percentage also indicated better effects than placebo (RR = 1.16, 95% CI 1.02 to 1.31, $p = 0.02$; RR = 1.28, 95% CI 1.07 to 1.54, $p = 0.008$; Fig. 3c and d). However, after excluding the study of deucravacitinib by Morand et al., there was no significant difference between the treatment groups and the placebo group in BICLA response or LLDAS. Most studies did not show a significant improvement in SLEDAI-2 K. The levels of anti-dsDNA titer and complements did not show significant changes after treatment compared with the placebo. The original data for this section are shown in Supplementary Table S6.

3.4. Effects of JAK inhibitors on organ involvement in SLE

Five RCTs reported on the effects of JAK inhibitors on the mucocutaneous and musculoskeletal systems. Baricitinib showed more marked effects based on SLEDAI-2 K remission of arthritis or rash, with an RR of 1.09 (95% CI 1.00 to 1.18; $p = 0.04$, Fig. 3e) compared to placebo. In the phase III study of baricitinib, researchers conducted a non-multiplicity-controlled analysis of improvement in some organ domains of interest and found significant improvement over placebo in both musculoskeletal and mucocutaneous domains.

Different doses of deucravacitinib were found to reduce Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) activity by >50% (CLASI-50) and were significantly better than placebo. Fourteen patients from case reports or series had mucocutaneous lesions at baseline. Three patients suffered from severe alopecia, and all were cured after starting JAK inhibitors. Twelve people had rash due to SLE, and after treatment with baricitinib or tofacitinib, nine people had rash resolutions (Table 1).

Organ-specific endpoints for arthritis were significantly improved with deucravacitinib 3 mg twice daily versus placebo (mean change from baseline in the active joint count, -8.9 versus -7.6 , respectively, $p = 0.001$). For the evaluation of arthritis, tender and swollen joint counts declined after being treated with baricitinib 4 mg in three RCTs. Two physicians encountered SLE patients complicated with Jaccoud's arthropathy (JA), and they treated them with upadacitinib and tofacitinib, respectively, and both patients' joint symptoms improved a few months later. In a case series, all four patients experienced alleviation of

Table 1
Characteristics of the included studies.

Author, year	Study type (Registration number)	Disease	Treatment regimen	Patient number	Primary outcome (RCTs)/ Clinical outcome (cases)	Time frame (weeks)	Level of evidence
Morand, Vital et al., 2023 [6]	RCT (NCT03616912)	SLE	Baricitinib 2 mg qd Baricitinib 4 mg qd Placebo	255 252 253	percentage of achieving SRI-4 response (4 mg baricitinib)	52	2
Petri et al., 2023 [7]	RCT (NCT03616964)	SLE	Baricitinib 2 mg qd Baricitinib 4 mg qd Placebo	261 258 256	percentage of achieving SRI-4 response (4 mg baricitinib)	52	2
Morand et al., 2023 [13]	RCT (NCT03252587)	SLE	Deucravacitinib 3 mg bid Deucravacitinib 6 mg bid Deucravacitinib 12 mg qd Placebo	91 93 89 90	percentage of achieving SRI-4 response	32	2
Hasni et al., 2021 [14]	RCT (NCT02535689)	SLE	Tofacitinib 5 mg bid Placebo	20 10	safety of tofacitinib in SLE	5 years	3
Wallace et al., 2018 [15]	RCT (NCT02708095)	SLE	Baricitinib 2 mg qd Baricitinib 4 mg qd Placebo	105 104 105	percentage of patients achieving remission of arthritis and/or rash as defined by the SLEDAI-2 K	24	2
Kahl et al., 2016 [16]	RCT (NCT01777256)	SLE	GSK2586184 50 mg bid GSK2586184 100 mg bid GSK2586184 200 mg bid GSK2586184 400 mg bid Placebo	9 10 10 10 11	change from baseline of SELENA SLEDAI score at indicated time points up to week 16	16	3
Baker et al., 2020 [17]	RCT (NCT03285711)	LN	Filgotinib 200 mg qd Lanraplenib 30 mg qd	5 4	percent change in urine protein	16	3
Werth et al., 2022 [18]	RCT (NCT03134222)	CLE	Filgotinib 200 mg qd Lanraplenib 30 mg qd Placebo	17 19 9	change in CLASI activity score	12	3
NCT03958955 [19]	RCT (NCT03958955)	CLE	Delgocitinib cream 2% bid Delgocitinib cream vehicle	27	target lesions with an IGA score of 0 or 1	6	3
Presto et al., 2018 [20]	RCT (NCT01597050)	CLE	Topical R333 6% bid Placebo	36 18	decrease in the total combined erythema and scaling score of all treated lesions	4	3
Yang et al., 2022 [21]	Case report	SLE	Tofacitinib 5 mg bid	1	(Refractory bullous SLE) Skin lesions were 50% improved in the second week and nearly 100% improved in the fourth week. SLE remained stable with a SLEDAI of 0 for 1 year.	1 year	5
Michailidou et al., 2022 [22]	Case report	SLE	Upadacitinib	1	(Jaccoud's arthropathy) Arthritis went into remission a few months later.	–	5
S Chen et al., 2022 [23]	Case report	SLE	Tofacitinib 5 mg bid	1	(Jaccoud's arthropathy) Joint symptoms improved. No further progression.	5 months	5
D Chen et al., 2022 [24]	Case report	SLE	Baricitinib 2 mg qd	1	(Patchy alopecia) Prominent hair regrowth and ESR decreased from 51 to 26 after 1.5 months. SLEDAI decreased from 4 to 0, and the PSL dose tapered to 5 mg/day after three months.	4.5 months	5
Chen et al., 2021 [25]	Case report	SLE	Tofacitinib 5 mg bid	1	(Long-standing refractory diffuse non-scarring alopecia and rash) The patient had noticeable hair regrowth after 4 weeks without any rash. MP was reduced to 8 mg/day with no relapse.	8 months	5
Maeshima et al., 2020 [26]	Case report	SLE	Baricitinib 4 mg qd	1	(Diffuse non-scarring alopecia) The patient had noticeable hair regrowth after 8 weeks. The PSL dose was reduced to 12.5 mg with no relapse.	12	5
Fornaro et al., 2020 [27]	Case report	SLE	Baricitinib 4 mg qd	1	(Refractory skin papulosquamous rash) Complete resolution of active skin lesions and stopped glucocorticoids after 4 weeks.	4	5
You et al., 2019 [28]	Case series	SLE	Tofacitinib 5 mg bid	10	4/4 had resolutions of arthritis. 6/9 had rash resolutions, 1 had alleviation, 1 unalleviated, and 1 flared.	1–12 months	4
Zhan et al., 2023 [29]	Case report	BLLE	Baricitinib 4 mg qd	1	The inflammation and scales of the lesions were significantly reduced.	8 months	5

(continued on next page)

Table 1 (continued)

Author, year	Study type (Registration number)	Disease	Treatment regimen	Patient number	Primary outcome (RCTs)/ Clinical outcome (cases)	Time frame (weeks)	Level of evidence
Maruyama et al., 2023 [30]	Case report	SCLE	Topical delgocitinib 0.5%	1	All rashes subsided after 2 weeks and no flares occurred for 6 months.	6 months	5
Park et al., 2022 [31]	Case report	DLE	Topical ruxolitinib 1.5%	1	Plaques showed improvement and subtle hair regrowth.	2 months	5
Kreuter et al., 2022 [32]	Case report	SCLE	Baricitinib 4 mg qd, 2 mg qd	1	The patient had complete clearance of SCLE, and the progression of the alopecia stopped.	–	5
Joos et al., 2022 [33]	Case report	SCLE	Baricitinib 4 mg qd	1	The patient had rapid improvement of the skin manifestations, and CLASI went from 21 to 3 after 6 months.	6 months	5
Calugareanu et al., 2022 [34]	Case report	DLE	Upadacitinib 15 mg/day	1	The CLASI-A score decreased by over 50% after 3 months.	3 months	5
Bonnardeaux et al., 2022 [35]	Case series	DLE, SCLE	Tofacitinib 5 mg bid	3	Patient 1: DLE improved after 4 months, CLASI from 28 to 8. Patient 2: Resolution of all SCLE lesions. Patient 3: Effectiveness on limbs but ongoing facial DLE, tofacitinib stopped.	4 months 7 months 1 month	4
Mazori et al., 2021 [36]	Case report	DLE	Topical tofacitinib 2% bid	1	Prednisone tapered to 5 mg/day rapidly. DLE started to clear within 2 weeks and nearly resolved within 4 weeks.	4	5
Elman et al., 2022 [37]	Case report	ChLE	Tofacitinib 5 mg bid	1	ChLE improved significantly after 8 weeks.	8	5
Zimmermann et al., 2019 [38]	Case series	ChLE	Baricitinib 4 mg/day	3	ChLE improved significantly after 3 months, with reduced episodes in the winter, improved perfusion of the digits, and complete or partial relief of the skin.	3 months	4
Briand et al., 2019 [39]	Case report	ChLE	Ruxolitinib 10-15 mg/day	1	Rapid clinical improvement in one week. Complete resolution of skin lesions after 12 months.	12 months	5
Wenzel et al., 2016 [40]	Case report	ChLE	Ruxolitinib 20 mg bid	1	Complete remission of all lesions within 4 months, remission had been stable for 3 more months.	7 months	5

RCT, randomized controlled trial; SLE, systemic lupus erythematosus; LN, lupus nephritis; CLE, cutaneous lupus erythematosus; BLLE, blaschko linear lupus erythematosus; SCLE, subacute cutaneous lupus erythematosus; DLE, discoid lupus erythematosus; ChLE, chilblain lupus erythematosus; qd, once a day; bid, twice a day; SRI-4, Systemic Lupus Erythematosus Responder Index-4; SLEDAI-2 K, Systemic Lupus Erythematosus Disease Activity Index 2000; SELENA-SLEDAI, Safety of Estrogens in Lupus Erythematosus National Assessment Version of the Systemic Lupus Erythematosus Disease Activity Index; IGA, Investigator's Global Assessment Scale; CLASI, Cutaneous Lupus Erythematosus Disease Area and Severity Index; CLASI-A, CLASI-activity; SLEDAI, Systemic Lupus Erythematosus Disease Activity Index; ESR, erythrocyte sedimentation rate; MP, methylprednisolone; PSL, prednisolone.

Study	Experimental	Comparator	D1	D2	D3	D4	D5	Overall	
Morand, Vital et al, 2023	Baricitinib	placebo	+	+	+	!	!	!	Low risk
Petri et al, 2023	Baricitinib	Placebo	+	+	+	!	!	!	Some concerns
Morand et al, 2023	Deucravacitinib	Placebo	+	+	+	+	!	!	High risk
Hasni et al, 2021	Tofacitinib	Placebo	+	+	+	+	!	!	
Wallace et al, 2018	Baricitinib	Placebo	+	+	+	!	!	!	D1 Randomisation process
Kahl et al, 2016	GSK2586184	Placebo	+	+	-	!	!	-	D2 Deviations from the intended interventions
Baker et al, 2020	Filgotinib	Lanraplenib	+	+	-	!	!	-	D3 Missing outcome data
Werth et al, 2022	Filgotinib	Placebo	+	+	+	!	!	!	D4 Measurement of the outcome
NCT03958955	Delgocitinib cream	Placebo	+	+	+	!	!	!	D5 Selection of the reported result
Presto et al, 2018	Topical R333	Placebo	!	!	+	+	!	!	

Fig. 2. Risk of bias in the included clinical trials. Assessment of bias risk in randomized controlled trials involves evaluation of five key domains: (1) randomization process, (2) deviations from intended interventions, (3) missing outcome data, (4) measurement of outcome, and (5) selection of reported results. After evaluating these domains, the overall risk of bias is determined. A low risk of bias is only established when all five domains exhibit low risk. If one or more domains exhibit some concerns or high risk, then the overall risk of bias is determined to be at some concerns or high risk.

their arthritis symptoms after tofacitinib treatment, which was assessed using SLEDAI-2 K.

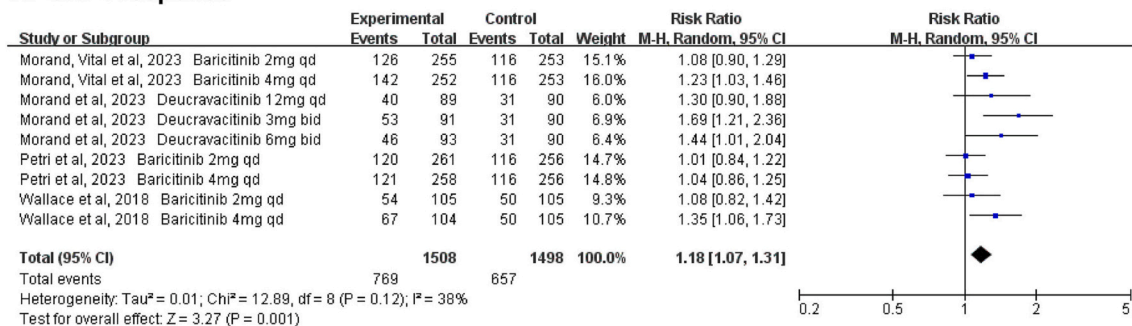
A phase II RCT evaluated the efficacy of filgotinib in patients with lupus membranous nephropathy compared with lanraplenib, a spleen tyrosine kinase inhibitor. There was a median reduction of 50.7% in 24-h urine protein and 51.6% in spot Urine Protein-to-Creatinine Ratio after 16 weeks of treatment with filgotinib, and two of four patients achieved

complete remission (defined as urine protein excretion below 0.5 g/day with no hematuria) in the filgotinib group.

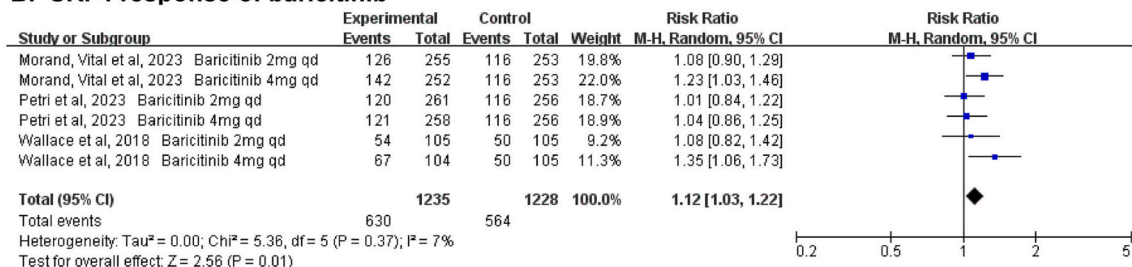
3.5. Effects of JAK inhibitors on CLE

Physicians used different orally and topically administered JAK inhibitors in treating cutaneous lupus erythematosus, including DLE,

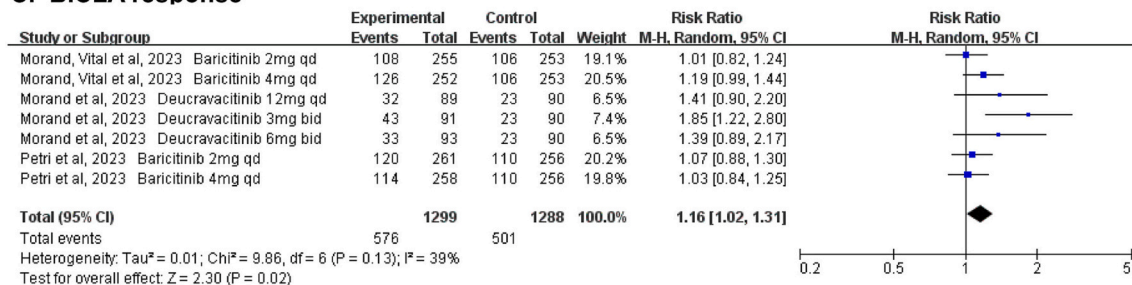
A. SRI-4 response



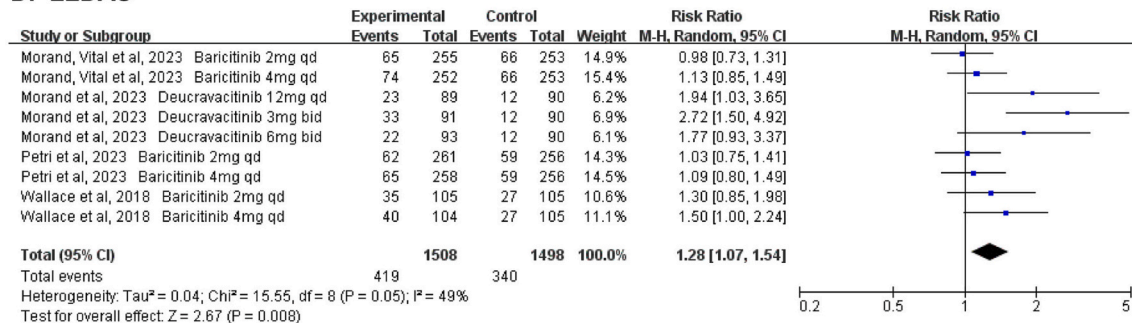
B. SRI-4 response of baricitinib



C. BICLA response



D. LLDAS



E. SLEDAI-2K remission of arthritis or rash

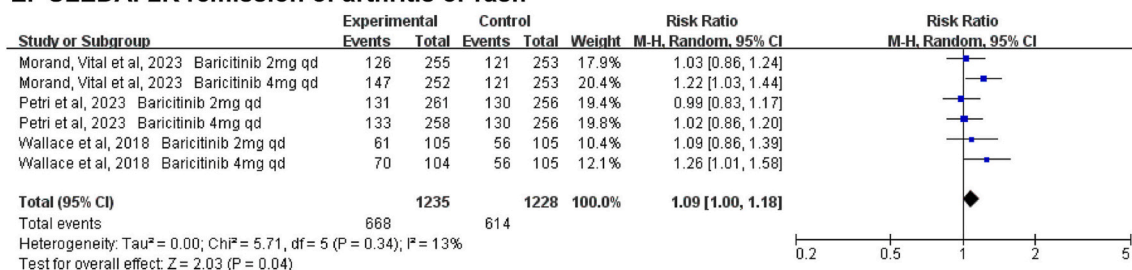


Fig. 3. Meta-analysis of the efficacy of Janus kinase inhibitors versus placebo. Relative risk of achieving SRI-4 response (A), SRI-4 response of baricitinib (B), BICLA response (C), LLDAS (D), and SLEDAI-2 K remission of arthritis or rash (E), compared JAK inhibitors with placebo.

SRI-4, Systemic Lupus Erythematosus Responder Index-4; BICLA, the British Isles Lupus Assessment Group (BILAG)-based Composite Lupus Assessment; LLDAS, Lupus Low Disease Activity State; SLEDAI-2K, Systemic Lupus Erythematosus Disease Activity Index 2000

SCLE, ChLE, and BLLE. Two phase II trials established the efficacy of topical delgocitinib and R333 in the treatment of DLE, but no primary or secondary outcomes were met. Another trial evaluated filgotinib in moderate to severe CLE and found that 50.0% and 68.8% of patients in the placebo and filgotinib arm, respectively, achieved a >5-point improvement in the CLASI-A score at week 12, but the primary endpoint (change from baseline in CLASI-A score at week 12) failed.

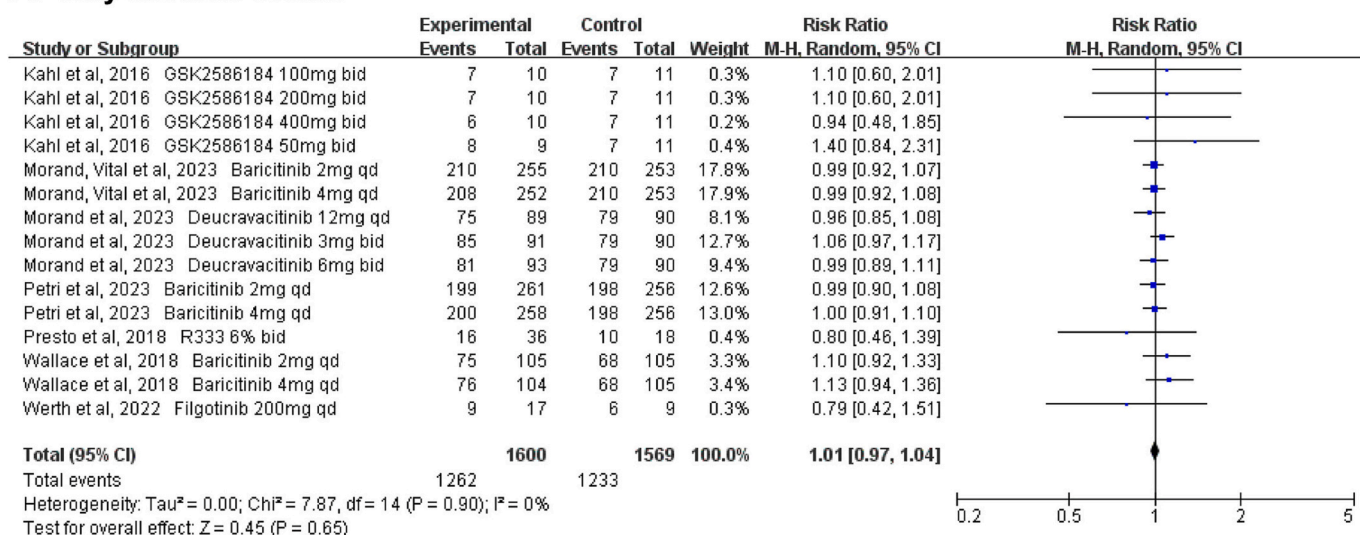
Sixteen patients with CLE were treated with JAK inhibitors and reported favorable effects (Table 1). Two patients complained of alopecia, and the patient treated with topical 1.5% ruxolitinib had subtle hair regrowth, while the other patient treated with baricitinib 4 mg/day had stopped the progression of the frontal fibrosing alopecia. Five patients suffering from DLE were treated with topically and orally administered JAK inhibitors, and four of them had an improvement in their skin lesions, while one patient using tofacitinib still had ongoing facial involvement and switched to a different treatment strategy. Among the four patients with SCLE, three of them had complete remission of the rash. Six ChLE patients were treated with baricitinib, ruxolitinib, and tofacitinib, which resulted in rapid clinical improvement within one to

eight months, and one patient treated with ruxolitinib 40 mg/day had a complete remission of all lesions.

3.6. Safety of JAK inhibitors

The incidence of adverse events (AEs) reported in all studies was low, and most of them were mild infections affecting the upper respiratory tract and urinary system. Overall, the JAK inhibitors group and the placebo group had a similar incidence of AEs (RR = 1.01; 95% CI 0.97 to 1.04; $p = 0.65$, Fig. 4a). The incidence of serious adverse events (SAEs) in the JAK inhibitor group was 10.2% (166 out of 1620 individuals), with no significant difference from the placebo group (RR = 1.27; 95% CI 0.98 to 1.66; $p = 0.07$). Sensitivity analyses showed the robustness of the synthesized results. As shown in Supplementary Fig S1f, the funnel plot for the analysis of any adverse events was symmetric, which indicated a small publication bias. In three RCTs, five venous thromboembolic events occurred in the baricitinib groups, and four patients treated with baricitinib and deucravacitinib had malignant tumors. In an RCT evaluating GSK2586184, a selective JAK1 inhibitor in SLE, SAEs were

A. Any adverse events



B. Serious adverse events

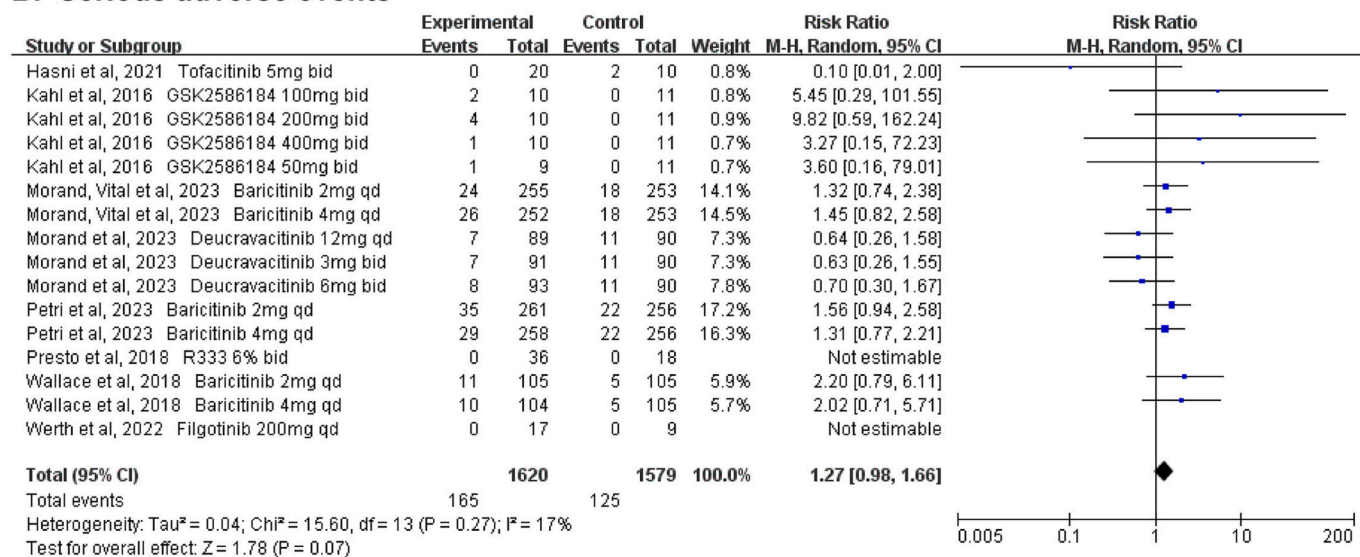


Fig. 4. Meta-analysis of the safety of Janus kinase inhibitors versus placebo. Relative risk of the incidence of any adverse events (A) and serious adverse events (B), compared JAK inhibitors with placebo.

reported in eight patients. Six patients had liver-related SAEs after initiation of GSK2586184. In four patients who experienced liver function test abnormalities, symptoms resolved upon treatment discontinuation. Two patients with symptoms consistent with DRESS syndrome were treated with high-dose steroids and then recovered.

4. Discussion

This is the first, most up-to-date systematic review summarizing the efficacy and safety of JAK inhibitors in treating SLE and CLE, incorporating the recently published results of two phase III RCTs of baricitinib for SLE (SLE-BRAVE-I and II) and one phase II RCT of deucravacitinib for SLE. Various JAK inhibitors were evaluated in RCTs, including baricitinib, tofacitinib, deucravacitinib, GSK2586184, filgotinib, topical delgocitinib, and topical R333. In addition, we included 20 case reports and series to observe real-world applications.

JAK is a family of intracellular, nonreceptor protein tyrosine kinases that interact with STAT to form the JAK-STAT signaling pathway [41]. Ligands bind to JAKs to initiate the pathway's activation, which primarily includes cytokines binding with type I or II cytokine receptors, such as interferons (IFNs), IL-6, IL-12, and IL-23 [42]. It can be logically deduced that via inhibition of JAKs, we can block specific intracellular signals stimulated by cytokines that are crucial in the pathogenesis of SLE and CLE. Furumoto Y and colleagues administered tofacitinib to MRL/lpr lupus-prone mice and observed that it ameliorated lupus and lupus-associated vascular dysfunction [43]. Fetter T and colleagues discovered that topical therapy with a specific JAK-1 inhibitor greatly improves CLE lesions in lupus-prone TREX1^{-/-} mice [44]. These preclinical evidences have led to the practice of JAK inhibitors treatment in SLE and CLE patients, which we have evaluated in our systematic review and meta-analysis.

Our meta-analysis of included RCTs found that baricitinib and deucravacitinib could decrease the activity of SLE, as evaluated by SRI-4 response, BICLA response, and LLDAS. However, the serological activity, including the levels of anti-dsDNA and complements, did not significantly improve. The JAK-STAT pathways involve various cytokines that influence the functions of B cells and plasma cells. For instance, IL-6 plays a critical role in promoting B cell differentiation and supporting the survival of plasmablasts, while IFN, IL-12, IL-23, and other cytokines induce B cell differentiation and enhance antibody production [45]. Although the studies included in our analysis did not reveal any significant decreases in autoantibodies, we remain hopeful that JAK inhibitors could inhibit antibody production. However, the effects of JAK inhibition on the complement pathway have not yet been explored, and thus, their impacts remain unclear.

Two recently published phase III clinical trials for baricitinib in SLE had a similar design, both using SRI-4 as the primary outcome. However, one trial (SLE-BRAVE-I) achieved the primary outcome, while the other (SLE-BRAVE-II) failed. The contradictory results highlight the need for advanced clinical trials. SLE is a heterogeneous disorder with multiple response domains, which can make it challenging to design effective trials and response criteria. Additionally, background treatment for SLE is typically at a high level, further complicating the evaluation of new treatments [46]. Although SRI-4 is the most frequently used primary outcome for evaluating the medications efficacy in SLE [47], some trials have explored other measurements. In a phase II RCT of baricitinib, resolution of arthritis or rash, as defined by the SLEDAI-2 K, was used and obtained a positive result, demonstrating the efficacy of baricitinib in treating skin and joint lesions. The phase III trial of anifrolumab in active SLE successfully met the BICLA response as the primary outcome [48], after a former trial using SRI-4 as the primary outcome had failed. The BICLA is a modification of the BILAG-2004, which can detect partial and complete improvement of an organ system. In contrast, SRI is unable to capture partial improvements and instead requires a full resolution within a specific item to register change [49,50]. We suggest that future clinical trials of JAK inhibitors in SLE prudently choose the primary

outcome of the study and highlight the advantages of JAK inhibitors in treating skin and joint lesions in the study design.

Our results illustrated that JAK inhibitors had a promising effect on the musculoskeletal and mucocutaneous systems. In two RCTs of baricitinib, remission of arthritis or rash as defined by SLEDAI-2 K significantly improved compared with placebo. In case reports and series, physicians tended to use JAK inhibitors to treat skin and joint lesions and made gratifying headway. These patients were previously treated with glucocorticoids at a dose above 10 mg prednisone (or equivalent), hydroxychloroquine (HCQ), and one or more immunosuppressants, but did not experience satisfactory effects. After switching to JAK inhibitors, glucocorticoids could be tapered, and their symptoms were alleviated. The efficacy on skin and joints was quite acceptable. Five kinds of JAK inhibitors have been approved for the treatment of rheumatoid arthritis, proving their efficacy in reducing joint inflammation. The theoretical foundation of their effects on skin involvement will be discussed in the CLE part. Only one RCT explored the application of JAK inhibitors in lupus nephritis, and since it did not compare JAK inhibitors with a placebo, we could not draw any conclusions.

All three RCTs that evaluated the efficacy of JAK inhibitors in CLE failed. Two of them used topical JAK inhibitors, which could only alleviate local inflammation, while the only RCT assessing oral filgotinib had a small sample size. However, case reports and series showed that a variety of JAK inhibitors could effectively treat DLE, SCLE, ChLE, and BLLE, resulting in remission in a short period of time. Dermatologists have already found that JAK inhibitors are efficacious in many skin diseases, such as alopecia areata, atopic dermatitis, and psoriasis [51]. Skin lesions in lupus are characterized by an overexpression of interferon-regulated cytokines and the presence of type I IFN-producing plasmacytoid dendritic cells and keratinocytes [52], which can be blocked by JAK inhibition. The etiology of non-scarring alopecia in SLE remains elusive. However, it has been posited that the involvement of hair follicles may share a common pathophysiology with CLE. Specifically, the inhibition of the anagen phase of hair growth and subsequent premature hair shedding are primarily driven by the cytokine IFN- γ , with emerging evidence suggesting that IFN- α may also exert an effect on hair follicles [53]. Notably, inhibiting the JAK-STAT pathway has been shown to mitigate the progression of IFN-induced hair loss in SLE.

Overall, JAK inhibitors are safe, with the most common adverse event being infection. In our analysis, there was no difference in the incidence of AEs and SAEs between JAK inhibitors and placebo. However, in an RCT of GSK2586184, six out of 39 patients had severe hepatic adverse events, which warranted attention and further investigation. Previous studies have shown that JAK inhibitors have similar safety profiles to other biological disease-modifying anti-rheumatic drugs (bDMARDs) [54–57]. Although the mechanisms of JAK inhibitors may indicate a higher risk of infection, cancer, and venous thrombosis [58–60], we did not observe any concerning trends in the current studies of SLE.

There are several ongoing clinical trials of JAK inhibitors in SLE and CLE. AbbVie has reported positive results for the phase II RCT of upadacitinib (RINVOQ) in moderate to severely active SLE. The trial met its primary endpoint of SRI-4 and achieved steroid dose reduction to <10 mg prednisone equivalent by week 24. No new safety signals were observed. As a result, AbbVie is advancing its upadacitinib program for SLE to Phase III [61]. Another phase II RCT is evaluating the efficacy and safety of deucravacitinib in DLE and/or SCLE [62]. In addition, there are two phase III studies for deucravacitinib in SLE [63,64], and one phase III study comparing baricitinib with cyclophosphamide in LN [65].

There were some limitations to this study. First, the evaluations of efficacy in various studies exhibit variability, and some trials demonstrate a high degree of bias, necessitating the selection of clinical trials for comprehensive analysis. Second, case reports and case series lacked quantitative data, making the assessment of efficacy relatively challenging. Additionally, reporting bias may have been an issue in these studies, as failed treatments may not have been reported.

5. Conclusion

In conclusion, our systematic review suggests that JAK inhibitors have the potential to be a viable treatment option for systemic and cutaneous lupus erythematosus, particularly in the treatment of cutaneous and musculoskeletal lesions associated with SLE. Because the impact of serological changes remains uncertain, there is a need for further investigation to confirm and expand upon these findings.

Author contributions

Leyao Ma and Liying Peng contributed equally to this manuscript.

Literature search: L Ma, L Peng, M Li; Study design: L Ma, L Peng, J Zhao, M Li; Data collection: L Ma, L Peng, S Zhang, C Wu; Data analysis: L Ma, L Peng, W Bai, Nan Jiang, W Zhang, Y Zhao, X Tian; Data interpretation: L Ma, L Peng, L Wang, D Xu, X Leng, Q Wang; Drafting of the manuscript: L Ma, L Peng, J Zhao, M Li; Critical revision: J Zhao, W Bai, N Jiang, S Zhang, C Wu, L Wang, D Xu, X Leng, Q Wang, W Zhang, Y Zhao, X Tian, M Li; Final approval of the version to be published: L Ma, L Peng, J Zhao, W Bai, N Jiang, S Zhang, C Wu, L Wang, D Xu, X Leng, Q Wang, W Zhang, Y Zhao, X Tian, M Li, Xiaofeng Zeng.

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Ethical approval

Not required for a systemic review.

Declaration of Competing Interest

The authors have declared no conflicts of interest.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.autrev.2023.103440>.

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